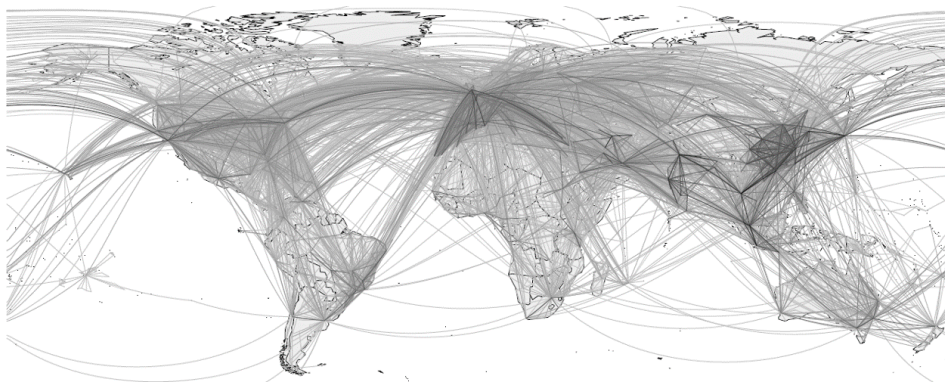


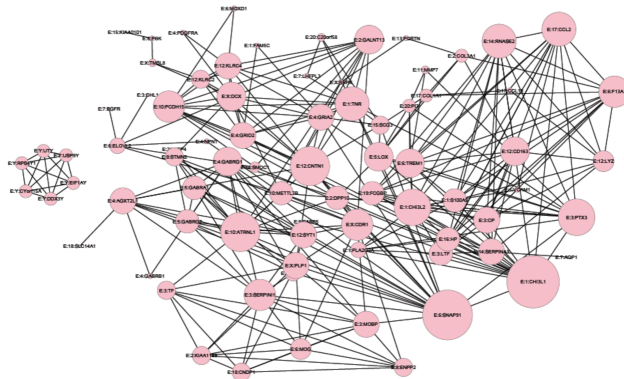
Unsupervised Learning: Graphical Models

Networks



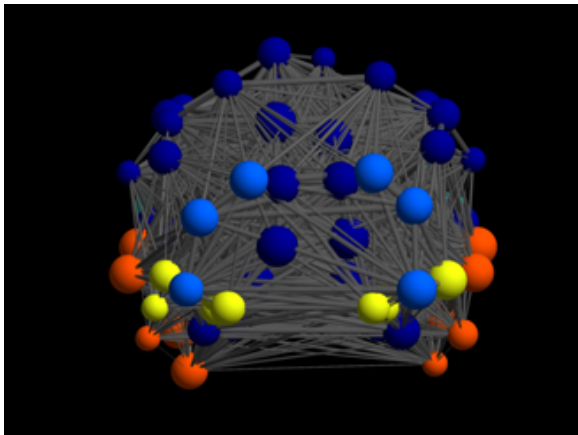
Depicts relationships (edges) between features (nodes).

Networks



Genomics: Relationships between genes.

Networks



Neuroimaging: Relationships between brain regions.

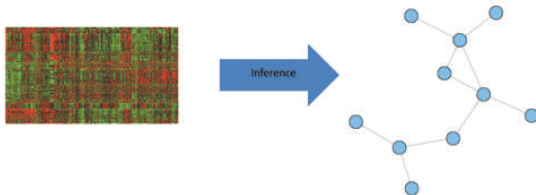
Networks in Unsupervised Learning

1 Network Data.

- ▶ Examples: Social networks, twitter, citations, surveillance, web links, etc.

2 **Our focus:** Learn network from data (structural network learning).

- ▶ Data matrix: $\mathbf{X}_{n \times p}$.
- ▶ Features form p nodes.
- ▶ Goal: Learn edges between nodes $\equiv$ learn the relationships between features.



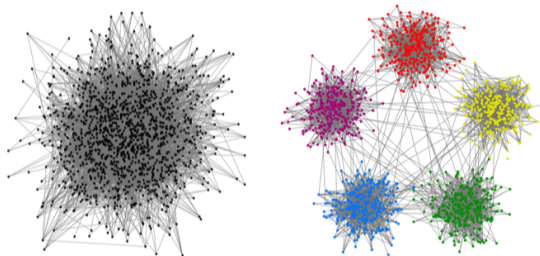
Network Data

- Network data have arisen as one of the most common forms of information collection.
- Networks consist of two components:
 - ① nodes or vertices corresponding to basic units of a system;
 - ② edges representing connections between nodes.
- Examples:
 - ① Nodes might be humans in social networks; molecules, genes, or neurons in biology networks, or web pages in information networks.
 - ② Edges could be friendships, alliances, URLs, or citations.
- A network can be represented by an adjacency matrix.

Network Data

Major Statistical Task: Community Detection

- Spectral Clustering.
- Modularity.
- Stochastic Block Models.
- Latent Variable Models.



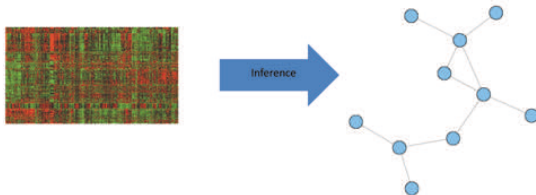
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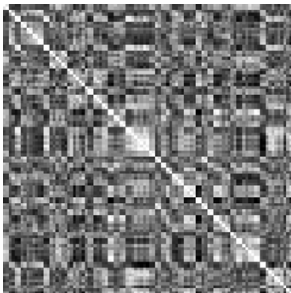
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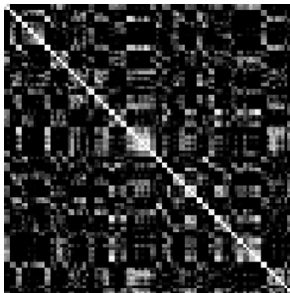


Correlation Networks (Association Networks)

- Simplest (and most-widely used!) method for estimating networks; assumes that edges correspond to large correlation magnitudes.
- Let $r(i, j)$ be correlation between X_i and X_j ; we claim an **edge between i and j** if $|r(i, j)| > \tau$.
 - ▶ τ : a user-specified threshold (**tuning parameter**).



Correlation matrix.



Thresholded correlation matrix.

Limitations of Correlation Networks

- ① The estimation is highly dependent on the **choice of τ** .
- ② Correlation captures **linear associations**, but **many real-world relationships are nonlinear**.
- ③ Large correlations can occur due to **confounding**.

Limitations of Correlation Networks

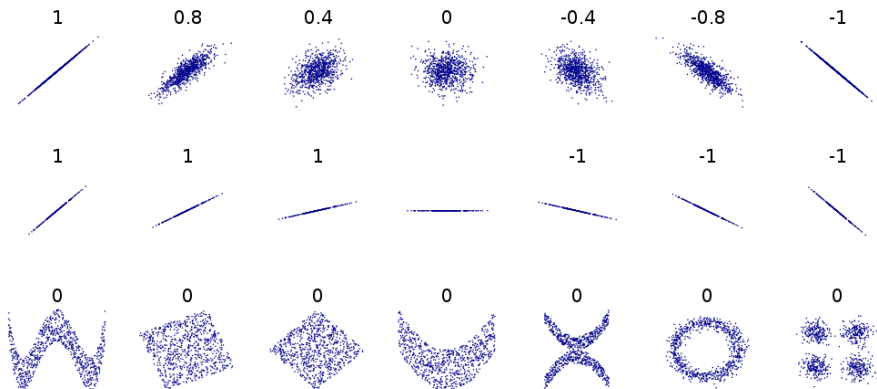
The estimation is highly dependent on the choice of τ .

- We can instead test $H_0 : r_{xy} = 0$
- A commonly used test is given by the Fisher transformation

$$Z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right) = \text{artanh}(r) \sim_{H_0} N \left(0, \frac{1}{\sqrt{n-3}} \right)$$

Limitations of Correlation Networks

Correlation captures **linear** associations, but **many real-world relationships are nonlinear**.



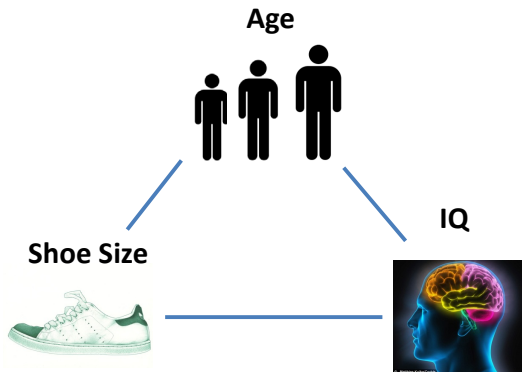
Limitations of Correlation Networks

Correlation captures **linear** associations, but **many real-world relationships are nonlinear**.

- We can use other measures of association, for instance, **Spearman correlation** or **Kendal's τ** .
 - ▶ These methods define correlation between two variables, based on the **ranking** of observations, and not their exact values.
 - ▶ They can better capture non-linear associations.
- We can instead use **mutual information**; this has been used in many algorithm, e.g. ARACNE.

Limitations of Correlation Networks

Large correlations can occur due to **confounding**.



Graphical Models

Probabilistic Graphical Models

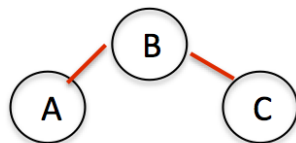
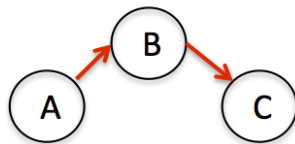
Joint multivariate probability distribution where dependencies can be represented as a network.

Advantages:

- Graphical models offer efficient factorized forms for joint distributions with easily interpretable dependencies.
 - ▶ **Conditional dependencies** denoted via an edge in network.
- Convenient visual representation.

Two Major Types of Graphical Models

- 1 Directed - Bayesian Networks.
 - ▶ Directed Acyclic Graphs.
- 2 Undirected - Markov Networks.
 - ▶ **Our Focus!**

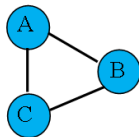


Markov Networks

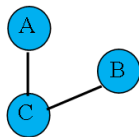
Markov Network

An *undirected graphical model* that characterizes **conditional dependence** ($\equiv$ direct relationships).

- *Edge*: Two nodes are **conditionally dependent**.
- *No edge*: Two nodes are **conditionally independent**.
- Conditions on all other nodes.



$$A \perp B \mid C$$

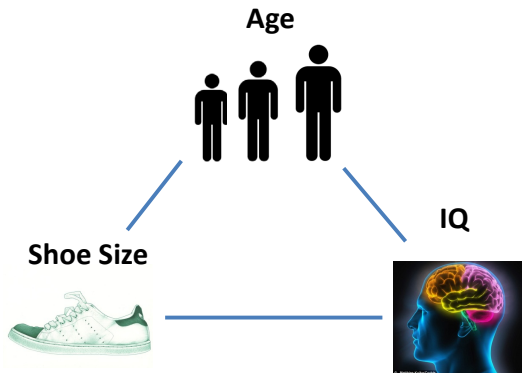


Markov Networks — Conditional Dependence

Regression Interpretation:

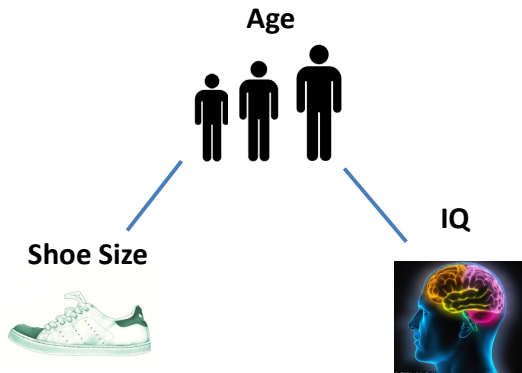
- Imagine trying to predict the observations in **Node A** (response) by the observations of all other nodes (predictors).
- **Node B** predictive of **Node A** (with all other nodes in model).
 - ▶ **A** is conditionally dependent on **B**.
 - ▶ Edge.
- Because of other nodes in model, **Node B** does not add any predictive value for **Node A**.
 - ▶ **A** is conditionally independent of **B**.
 - ▶ No Edge.

Markov Networks — Conditional Dependence



Correlation.

Markov Networks — Conditional Dependence



Conditional Dependence.

Markov Networks — Conditional Dependence

How can we learn conditional dependencies?

- A and B are conditionally independent given C if

$$P(A, B \mid C) = P(A \mid C)P(B \mid C)$$

- ▶ Generally difficult (need to estimate multivariate densities).
- Alternatively, can use nonparametric approaches, e.g. **conditional mutual information**, but not easy in high dimensions.
- Often resort to models, or simple measures, such as **partial correlations**...

Partial Correlation

- Partial correlation measures the correlation between A and B after the effect of the other variables are removed.
 - In our example, this means correlation between shoe size and IQ, after adjusting for age.
- When C is a single variable, the partial correlation between A and B given C is given by:

$$\rho_{AB.C} \equiv \rho(A, B|C) = \frac{\rho_{AB} - \rho_{AC}\rho_{BC}}{\sqrt{1 - \rho_{AC}^2}\sqrt{1 - \rho_{BC}^2}}.$$

- Alternatively, regress A on C and get the residual, r_A ; do the same for B to get r_B . The partial correlation between A and B given C is $\text{Cor}(r_A, r_B)$.
- In general multivariate settings, partial correlation between X_i and X_j can also be computed from the inverse of the covariance matrix Σ : $(\Sigma^{-1})_{i,j} / \sqrt{(\Sigma^{-1})_{i,i}(\Sigma^{-1})_{j,j}}$.

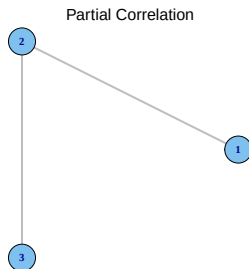
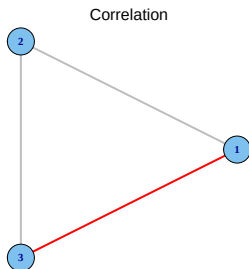
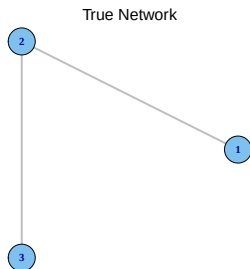
Partial Correlation

- Partial correlation is **symmetric** $\Rightarrow$ **undirected network**
- Partial correlation is a number **between -1 and 1**
- In partial correlation networks, we **draw an edge** between A and B , **if the partial correlation between them is large**
- Calculation of partial correlation is more difficult in high dimensions

A Simple Example

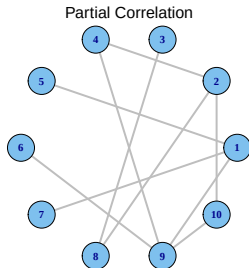
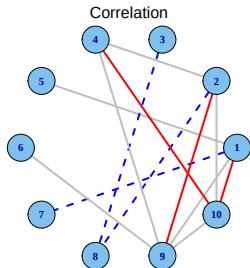
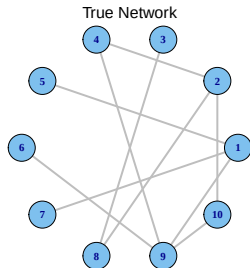
$$\text{Correlation} = \begin{bmatrix} 1 & .75 & .5625 \\ .75 & 1 & .75 \\ .5625 & .75 & 1 \end{bmatrix}$$

$$\text{Partial Corr} = \begin{bmatrix} 1 & .6 & 0 \\ .6 & 1 & .6 \\ 0 & .6 & 1 \end{bmatrix}$$



A Larger Example

- A network with **10 nodes** and **20 edges**
- $n = 100$ observations
- Estimation using correlation & partial correlation (**20 edges**)

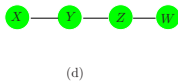
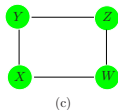
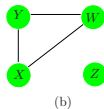
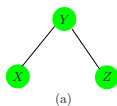


Gaussian Graphical Models (GGMs)

Partial Correlation for Gaussian Random Variables

- For Gaussian (multivariate normal) random variables, non-zero partial correlation is equivalent to conditional dependence relationship
- Reminder: Partial correlation between X_i and X_j given all other variables is given by the inverse of the (standardized) covariance matrix Σ .
 - ▶ The (i, j) entry in Σ^{-1} gives the partial correlation between X_i and X_j given all other variables $X_{\setminus i, j}$.
- Gaussian Graphical Model (GGM):
 - ▶ Multivariate normal: $\mathbf{X} \sim N(0, \Sigma)$
 - ▶ $\Theta = \Sigma^{-1}$ = inverse covariance/precision/concentration matrix.
 - ▶ Zeros in $\Theta \implies$ conditional independence!
 - ▶ Edges correspond to non-zeros in Θ .

Partial Correlation for Gaussian Random Variables



$$\begin{pmatrix} - & \times & 0 \\ \times & - & \times \\ 0 & \times & - \end{pmatrix}$$

$$\begin{pmatrix} - & \times & \times & 0 \\ \times & - & \times & 0 \\ \times & \times & - & 0 \\ 0 & 0 & 0 & - \end{pmatrix}$$

$$\begin{pmatrix} - & \times & 0 & \times \\ \times & - & \times & 0 \\ 0 & \times & - & \times \\ \times & 0 & \times & - \end{pmatrix}$$

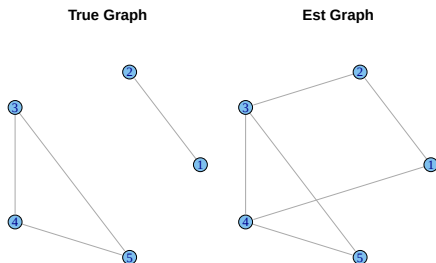
$$\begin{pmatrix} - & 0 & 0 & \times \\ 0 & - & \times & 0 \\ 0 & \times & - & \times \\ \times & 0 & \times & - \end{pmatrix}$$

Estimating GGMs

From our discussions so far, to estimate the network, we can

- 1 Calculate the **empirical covariance matrix**: for (centered) $n \times p$ data matrix X , $\mathbf{S} = (n - 1)^{-1} X^\top X$.
- 2 Find the inverse of $\mathbf{S}$. Non-zero values of $\mathbf{S}^{-1}$ give the edges.

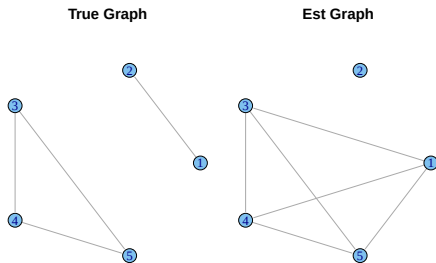
While simple, this may not work well in practice, even with large samples!



Estimating GGMs in High Dimensions

Many problems arise in high-dimensional settings, when $p \gg n$.

- First, **S is not invertible if $p > n$!**
- Even if $p < n$, but n is not very large, we may still get poor estimates, and many false positives/negatives.



Estimating GGMs in High Dimensions

- A number of methods have been proposed in the literature for estimating GGMs in high dimensions.
- The main idea in most of these methods is to use a regularization penalty, like the lasso.
- We discuss two approaches:
 - ▶ neighborhood selection
 - ▶ graphical lasso

Estimating GGMs in High Dimensions – Method 1

The idea behind **neighborhood selection**, is to estimate the graph by fitting a **penalized regression of each variable on all other variables**.

- Find **neighbors** of each node X_j by l_1 -penalized regression or lasso:

$$\underset{\beta^j}{\text{minimize}} \quad \|X_j - \mathbf{X}_{\neq j}\beta^j\|_2^2 + \lambda \sum_{k \neq j} |\beta_k^j|$$

- The final estimate is found by combining all of the edges from these individual regression problems.
 - ▶ Symmetry — β_k^j not always same as β_j^k .
 - ▶ Use min or max rule.

Estimating GGMs in High Dimensions – Method 2

Estimate a sparse Θ via penalized maximum likelihood estimation (MLE).

Graphical Lasso (glasso)

$$\underset{\Theta \succeq 0}{\text{maximize}} \quad \text{logdet}(\Theta) - \text{tr}(S\Theta) - \lambda \|\Theta\|_1$$

- **Blue**: Log-likelihood; logdet is log-determinant and tr is matrix trace.
- **Red**: Penalty that encourages zeros in off-diagonal elements of Θ .

Comparing the Two Approaches

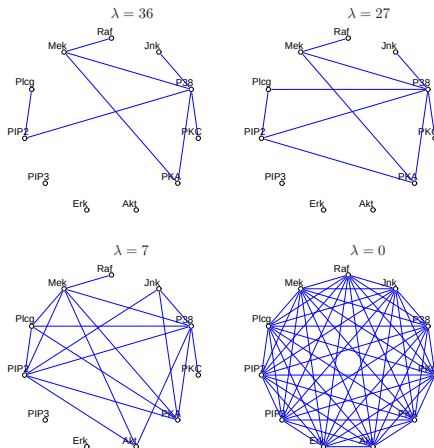
- Neighborhood selection is an **approximation to the graphical lasso**:
 - ▶ Consider regression of X_j on $X_k, j \neq k$
 - ▶ Then the regression coefficient for neighborhood selection is related to the j, k element of Θ :

$$\beta_k^j = -\frac{\Theta_{jk}}{\Theta_{jj}}$$

- Neighborhood selection is computationally more efficient, and may give better graph estimates, but does not give an estimate of Θ !

A Real Example

- **Flow cytometry** proteomics in single cells (Sachs et al, 2003).
- $p = 11$ proteins measured in $n = 7466$ cells



How to Choose λ ?

- λ modulates trade-off between **model fit** and **network sparsity**:
 - ▶ $\lambda = 0$ gives a dense network (no sparsity).
 - ▶ As λ increases, network becomes more sparse.

- A number of approaches available

- ① **Cross-Validation** — tends to yield overly dense networks.
- ② **Extended BIC** — adjusted BIC for high dimensions.
- ③ **Controlling the probability of falsely connecting disconnected components** at level α (Banerjee et al, 2008):

$$\lambda(\alpha) = \frac{t_{n-2}(\alpha/2p^2)}{\sqrt{n-2 + t_{n-2}(\alpha/2p^2)}},$$

($t_{n-2}(\alpha)$ is the $(100 - \alpha)\%$ quantile of t -distribution with $n - 2$ d.f.)

- ④ **Stability selection** — Choose λ that gives the most **stable network** (R: huge package)

Other Types of Graphical Models

Nonparanormal (Gaussian Copula) Models

- Suppose $X \approx N(0, \Sigma)$, but there exists **monotone functions** $f_j, j = 1, \dots, p$ such that $[f_1(X_1), \dots, f_p(X_p)] \sim N(0, \Sigma)$
 - ▶ X has a nonparanormal distribution $X \sim NPN_p(f, \Sigma)$.
 - ▶ f and Σ are parameters of the distribution, and estimated from data.
 - ▶ For continuous distributions, the nonparanormal family is **the same as the Gaussian copula family**
- One strategy to estimate the nonparanormal network:
 - Estimate the monotone functions:** $\hat{f}_j, j = 1, \dots, p$
 - transform the data:** $[\hat{f}_1(X_1), \dots, \hat{f}_p(X_p)]$
 - estimate the network of the transformed data** (e.g. calculate the empirical covariance matrix of the transformed data, and apply glasso or neighborhood selection)

A Related Procedure

- Liu et al (2012) and Xue & Zou (2012) proposed a closely related idea using **rank-based correlation**
 - ▶ Let r_j^i be the **rank of x_j^i** among $x_j^1, \dots, x_j^n$ and $\bar{r}_j = (n+1)/2$ be the average rank
 - ▶ Calculate **Spearman's ρ** or **Kendall's τ**

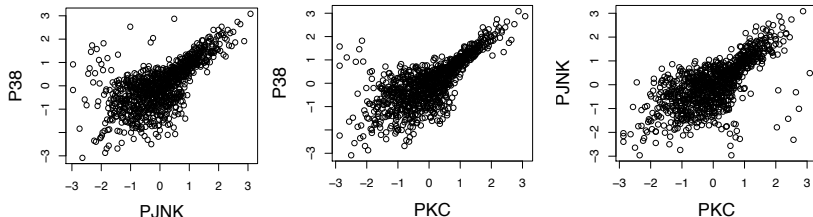
$$\hat{\rho}_{jk} = \frac{\sum_{i=1}^n (r_j^i - \bar{r}_j)(r_k^i - \bar{r}_k)}{\sqrt{\sum_{i=1}^n (r_j^i - \bar{r}_j)^2 \sum_{i=1}^n (r_k^i - \bar{r}_k)^2}}$$

$$\hat{\tau}_{jk} = \frac{2}{n(n-1)} \sum_{1 \leq i < i' \leq n} \text{sign} \left((x_j^i - x_j^{i'})(x_k^i - x_k^{i'}) \right)$$

- If $X \sim NPN_p(f, \Sigma)$, then $\Sigma_{jk} = 2 \sin(\rho_{jk}\pi/6) = \sin(\tau_{jk}\pi/2)$
- Therefore, we can estimate Σ^{-1} by **plugging in rank-based correlations into graphical lasso** (R-package huge)

A Real Data Example

- Protein cytometry data for cell signaling (Sachs et al, 2005)
- Transform the data using **Gaussian copula** (Liu et al, 2009), giving marginal normality
- Pairwise relationships still seem **non-linear**



- Shapiro-Wilk test rejects multivariate normality: $p < 2 \times 10^{-16}$

Graphical Models for Discrete Random Variables

- In many cases, biological data are not Gaussian: SNPs, RNAseq, etc
- Need to estimate CIG for other distributions: **binomial**, **poisson**, etc
- In this case, the estimators do not have a closed-form!
- A special case, which is computationally more tractable, is the class of **pairwise MRFs**

Pairwise Markov Random Fields

- The idea of **pairwise MRFs** is to “assume” that **only two-way interactions among variables** exist
 - ▶ The pairwise MRF associated with the graph G over the random vector X is the family of probability distributions $P(X)$ that can be written as

$$P(X) \propto \exp \sum_{(j,k) \in E} \phi_{jk}(x_j, x_k)$$

- ▶ For each edge $(j, k) \in E$, ϕ_{jk} is called the **edge potential function**
- For discrete random variables, any MRF can be transformed to an MRF with pairwise interactions by introducing additional variables¹

¹Wainwright & Jordan, 2008

Graphical Models for Binary Random Variables

- Suppose $X_1, \dots, X_p$ are binary random variables, e.g. corresponding to SNPs, or DNA methylation
- A special case of discrete graphical models is the **Ising model for binary random variables**

$$P_{\theta}(x) = \frac{1}{Z(\theta)} \exp \left\{ \sum_{(j,k) \in E} \theta_{jk} x_j x_k \right\}$$

- ▶ A **pairwise MRF** for binary data, with $\phi_{jk}(x_j, x_k) = \theta_{jk} x_j x_k$
- ▶ $x^i \in \{-1, +1\}^p$
- ▶ The **partition function** $Z(\theta)$ ensures that distribution sums to 1
- ▶ $(j, k) \in E$ iff $\theta_{jk} \neq 0$!

Graphical Models for Binary Random Variables

- We can consider a **neighborhood selection**² approach with an ℓ_1 penalty to find the neighborhood of each node

$$N(j) = \{k \in V : (j, k) \in E\}$$

- For $j = 1, \dots, p$, need to solve (after some algebra)

$$\min_{\theta} \left\{ n^{-1} \sum_{i=1}^n \left[f(\theta; x^i) - \sum_{k \neq j} \theta_{jk} x_j^i x_k^i + \lambda \|\theta_{-j}\|_1 \right] \right\}$$

$$\triangleright f(\theta; x) = \log \left\{ \exp \left(\sum_{k \neq j} \theta_{jk} x_k \right) + \exp \left(- \sum_{k \in -j} \theta_{jk} x_k \right) \right\}$$

- This is equivalent to **solving p penalized logistic regression** problems, which is straightforward (R-package `glmnet`)

²Ravikumar et al (2010)

Other Non-Gaussian Distributions

- Assume a **pairwise graphical model**

$$P(X) \propto \exp \left\{ \sum_{j \in V} \theta_j \phi_j(X_j) + \sum_{(j,k) \in E} \theta_{jk} \phi_{jk}(X_j, X_k) \right\}$$

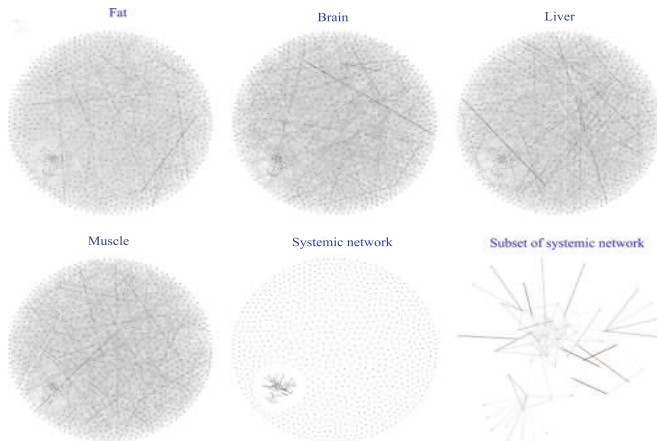
- Then, similar to the Ising model, graphical models can be learned for other members of the **exponential family**
 - ▶ Poisson graphical models (for e.g. RNAseq), Multinomial graphical models, etc
 - ▶ All of these can be learned using a **neighborhood selection approach**, using the `glmnet` package³
 - ▶ We can even learn networks with multiple types of nodes (gene expression, SNPs, and CNVs)⁴

³Yang et al (2012)

⁴Yang et al (2014), Chen et al (2015)

Research Highlight

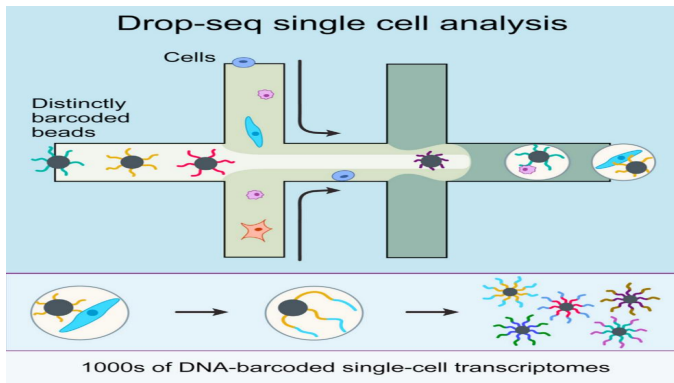
Multiple Dependent Gaussian Graphical Models



(Xie, Liu, Valdar, 2016)

Research Highlight

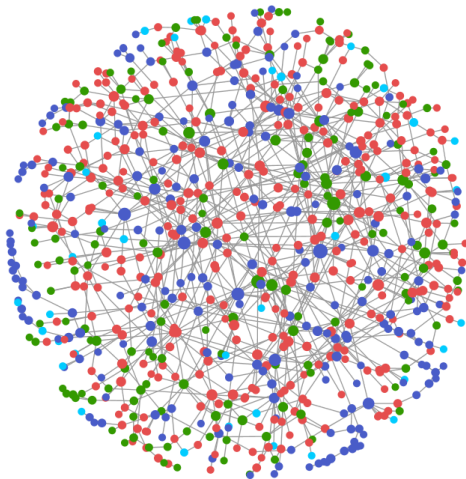
Graphical Models for Single Cell RNA-seq Data: count data with inflation of 0 and cell dependence.



Dependent Poisson and Hurdle graphical models (Liu et al., 2022)

Research Highlight

Mixed Graphical Models

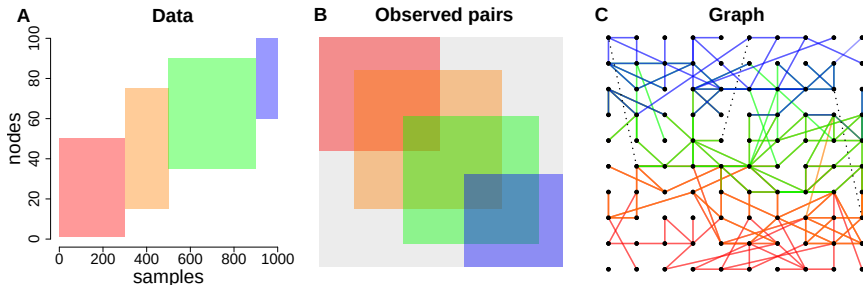


(Yang et al., 2014)

Research Highlight

Graph Quilting

Problem: How do you learn graphs from unobserved pairs of variables?



(Vinci, Dasarathy, and Allen, 2022)